Intro to Machine Learning

Major Hw1

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Part 1

Q1

the dataset has 1250 rows and 26 columns

Q2

In the real world, this feature refers to the number of conversations each patient had on average a day.

This feature is ordinal since it's values don't exceed 22, and the feature has a natural order, thus its an ordinal feature.

conversations_per_day	count
1	104
2	215
3	224
4	190
5	156
6	111
7	60
8	72
9	39
10	23
11	19
12	12
13	9
14	6
15	2
16	2
17	4

Q3

Feature Name	Description	Type
Patient_id	The ID of the patient	Other
Age	The age of the patient	Ordinal
Sex	The sex of the patient	Categorical
Weight	The weight of the patient	Continuous
Blood_type	The blood type of the patient	Categorical
Current_location	The current coordinates of the patient	Other
Num_of_siblings	The number of the siblings for the patient	Ordinal
Happiness_score	The happiness score for the patient - (out of 10)	Ordinal
Household_income	Number of people in the patient's household that have income.	Ordinal
Conversation_per_day	Number of conversations on average that the patient has a day	Ordinal
Sugar_levels	Sugar Levels for the patient	Continuous
Sport_activity	Weekly average of hours spent on sport activities the patient does.	Ordinal
Symptoms	The symptoms of the patient	Other
Pcr_date	The date that the patient had taken the PCR test	Ordinal
PCR_01	PCR test part 1 values	Continuous
PCR_02	PCR test part 2 values	Continuous
PCR_03	PCR test part 3 values	Continuous
PCR_04	PCR test part 4 values	Continuous
PCR_05	PCR test part 5 values	Continuous
PCR_06	PCR test part 6 values	Continuous
PCR_07	PCR test part 7 values	Continuous
PCR_08	PCR test part 8 values	Continuous
PCR_09	PCR test part 9 values	Continuous
PCR_10	PCR test part 10 values	Continuous

Q4

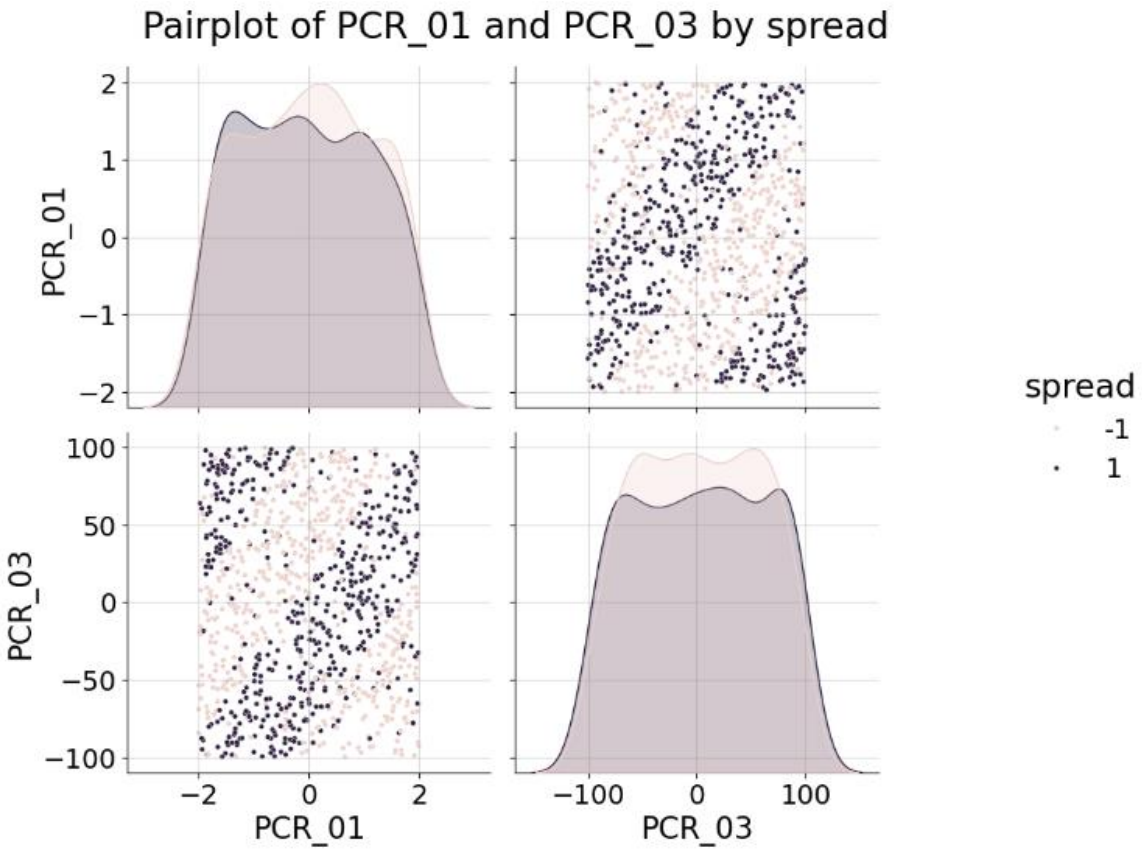
The decisions that we make is based on the training set that we have split, and if we don't provide our model with the same training set, decision from the past that we have made might not be relevant now.

When comparing between models' performances we should provide all models with the same training set, as the training has effect on the evaluation and thus giving different training sets to our models might be unfair and could introduce bias to the comparison.

Part 2

Q5

We can see that these features are useful for predicting the spread, as the datapoints are pretty much separable when looking at these two features (aside from a couple anomalies).



Q6

We can see that there is no correlation between each of the features PCR_01, PCR_03 with spread, this result doesn't contradict with our answer from the previous question, as the separability was a result of taking both features against spread where as the correlation took each feature on its own and compared it to spread, which as we can see doesn't give much.

```
Correlation between PCR_01 and spread is: -0.019
Correlation between PCR_03 and spread is: 0.006
```

Q7

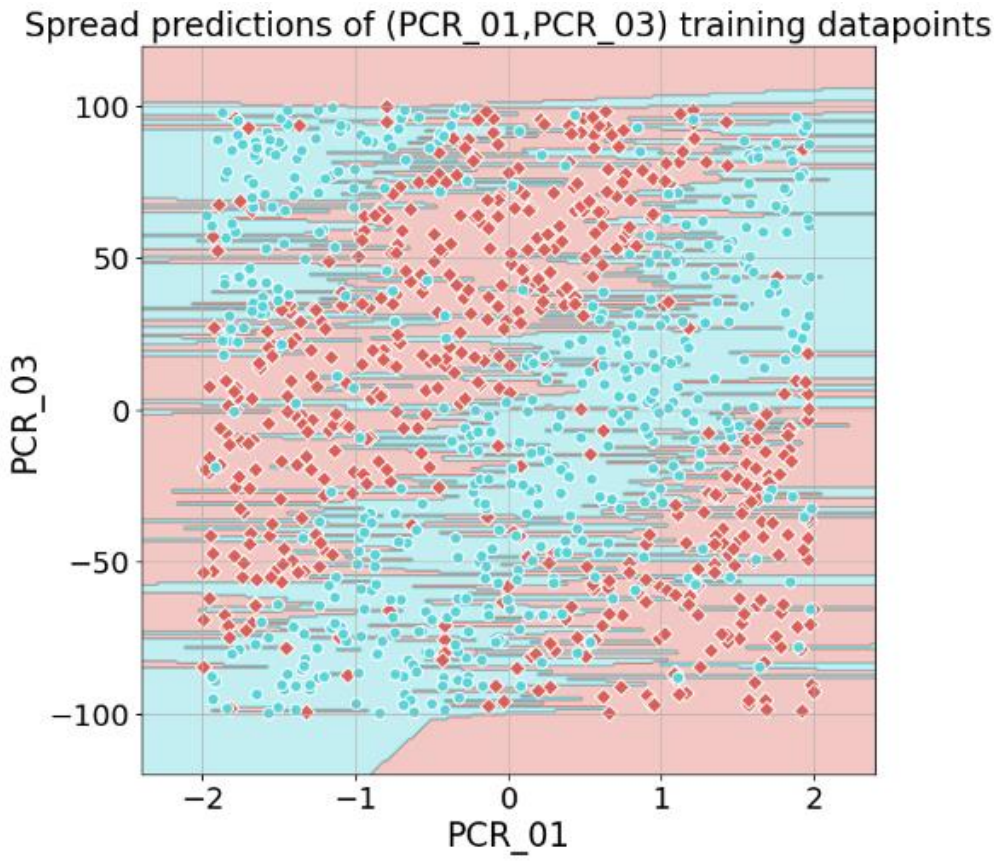
The complexity of our prediction function given a single test datapoint:

1. Calculating the distance between each of our training datapoint and the given test point:
   we use the scipy...csdist function to get the required distances, each of our points are d-dimensional and we have n training points.
   which means that we get time complexity of $O(nd)$
2. Find the indexes of k - nearest neighbors to our test point:
   for this we use the np.argpartition function which sorts each row and then takes the k lowest distances.
   we have n distances thus the sort takes $O(nlogn)$, and then we find the indexes of the k lowest distances which takes O(k).
   in conclusion this step takes: $O(nlogn + k)$.
3. Find the prediction by the majority labels of the K-NN:
   we use the np.bincount to get how many labels of the k-nearest neighbors are +1/-1 and then we take argmax() to find what our prediction should be for the given test point, so this takes us $O(k)$

In conclusion the prediction for a single test point takes: $O(nd + nlogn + k)$

Q8

- Training accuracy: 1.0
- Test accuracy: 0.68



Q9

Normalization is important in knn (and in most distance based models) because we predict the label for a given datapoint by its distance from other points, so if our features have different scale we might give higher weight to one feature that has a larger scale and thus when calculating the distance between points we take into consideration (almost) only the feature with the large scale.

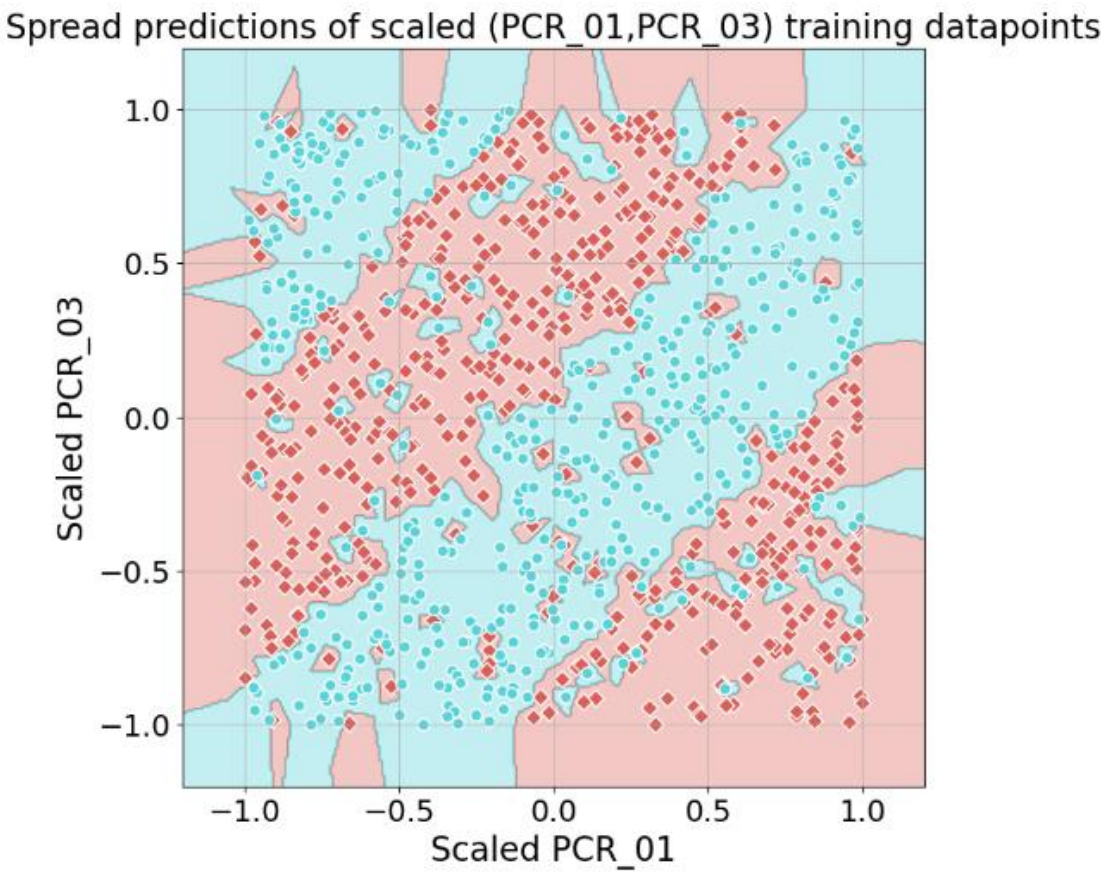
Formally, in our case the distance between 2 dim points are calculated by:

$$dist(A,B) = \sqrt{\left(A_{PCR_01} - B_{PCR_01}\right)^2 + \left(A_{PCR_03} - B_{PCR_03}\right)^2}$$

As we can see from the formula above if the PCR_01 feature has a higher scale than the PCR_03 feature the effect of $\left(A_{PCR_03} - B_{PCR_03}\right)^2$ will be neglected in calculating the distance.

So when we normalize our data we give the same weight to each feature which gives us better evaluations as we can see in the results below.

- Training accuracy: 1.0
- Test accuracy: 0.796

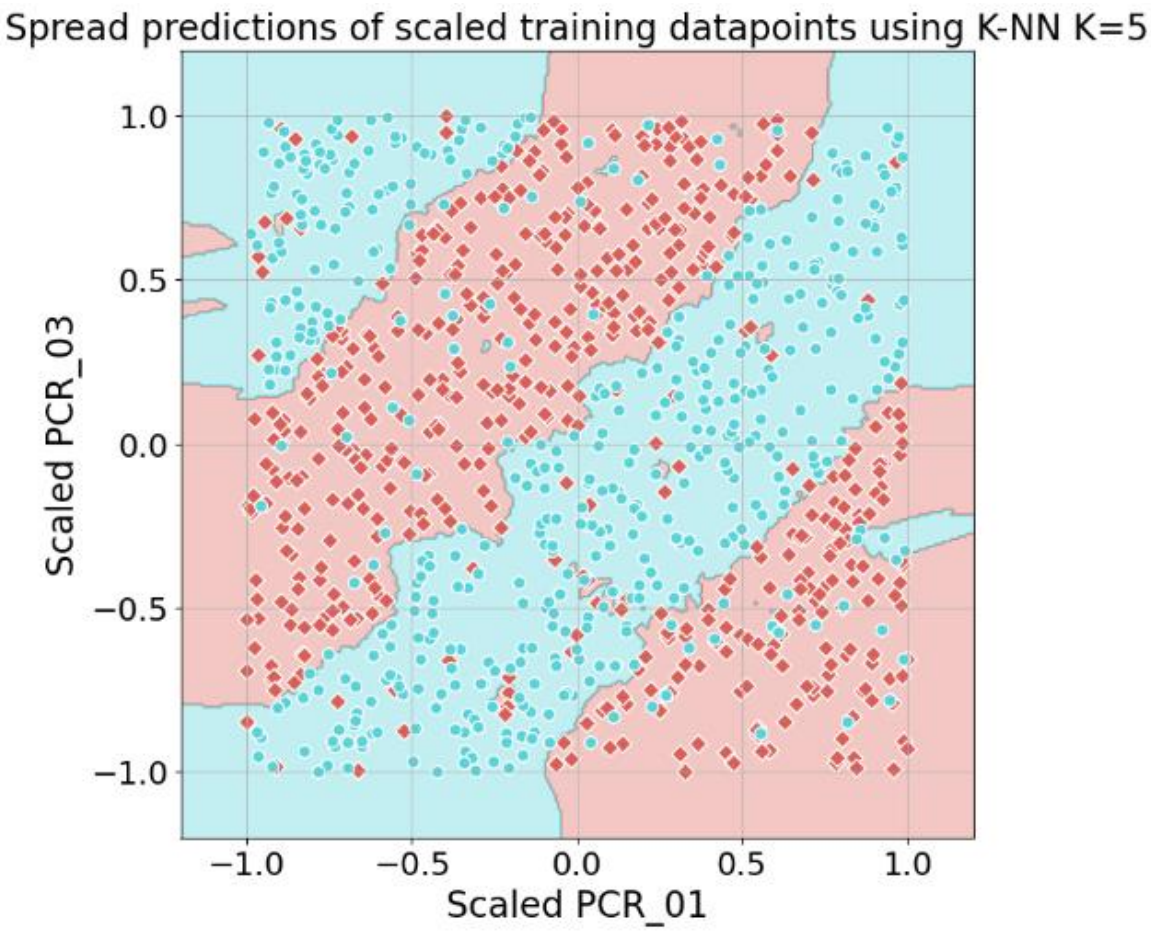


Q10

The value of K in the K-NN model decides the level of complexity that the classifier has, a lower value of K for example K=1 may cause overfitting and might cause the classifier to capture noise in the data, while a larger k for example K=5 might cause our classifier to give smoother decision boundary and makes the model less sensitive to individual data points (that may be noise).

We can see that in our case picking K=5 gave us a better test accuracy over K=1 and a smoother decision boundary as can be seen below, although we took a hit on our training accuracy that might be cause by noisy data points.

- Training accuracy: 0.886
- Test accuracy: 0.844



Q11

Normalizing the first feature that is uniformly distributed on the range [2,5] might not have a significant impact on our classifier, but when normalizing our second feature which is a chi-squared distribution that can have basically any number could potentially hurt our evaluation.

The second feature with chi-square distributed could include very large values (in low probability) and then when we apply the min-max normalization we'll get that most values are concentrated around 0, neglecting the actual distance between these points when using a distance based model like K-NN, As a result, the chi-squared distributed feature may not contribute meaningfully to the model's decision-making.

"In summary, blindly applying min-max scaling to features from different distributions, especially when one has a different nature than the other, can introduce biases in kNN models." – chatgpt

Part 3

Q12

We have 8 blood types ($\pm A, \pm B, \pm AB, \pm O$) thus we'll need 8 Boolean features to create a one-hot-encoding representation of the blood_type feature.

Q13

Since the symptoms feature holds multiple categorical values per entry, we created a new feature for each value (symptom). Each one of those new features gets a boolean value (0 or 1), 0 means that the patient does not have this symptom and 1 means that he has it.

Task E

The features that we classified as "other" are: patient_id, current_location and symptoms. There is no meaningful information in the patient_id's values for the target labels and therefore will not be of use to us in the learning process.

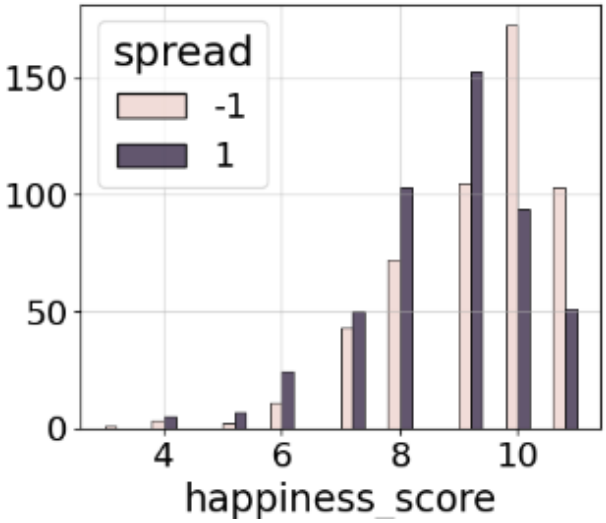
On the other hand, current location can point to contagious areas and hence be of use. The same applies to the Symptoms feature as we explained in a previous question.

As for Categorical we have sex and blood_type which are both useful features and we'll transform them both to numeric representation.

We'll also remove pcr_date as it doesn't affect our predictions.

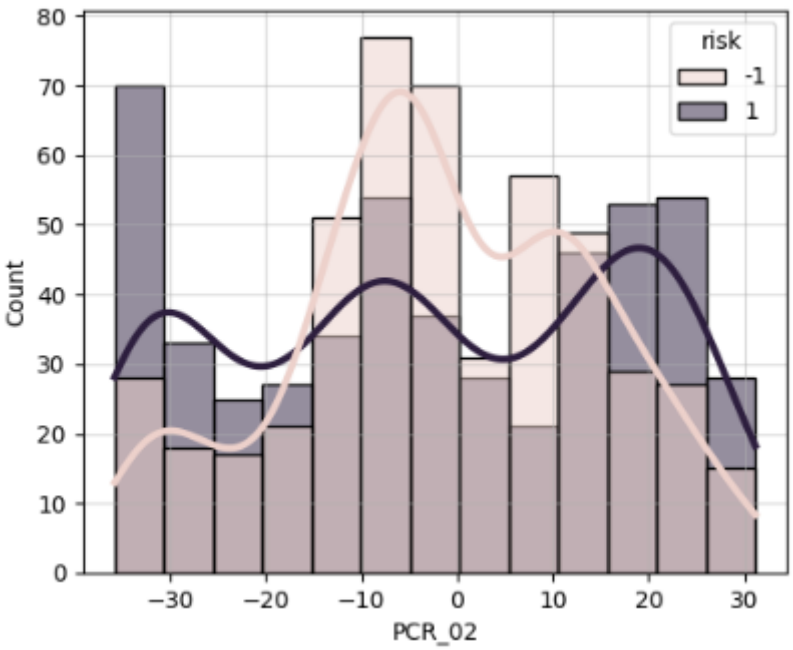
Q14

The happiness_score feature seems to be informative for predicting the spread target variable, from the graph below we can see a slight correlation between the happiness score and the chances of spreading the virus, in the sense that if the happy score is high were more likely to not spread the virus than we are to spread it.



Q15

The PCR_02 feature seems to be informative for predicting the risk target variable, from the graph below we can see a correlation between the PCR_02 values and whether there's risk, in the sense that if the PCR_02 values are around 0 (in the range [-10,10]) your likely to not be in risk, otherwise your likely to be in risk.

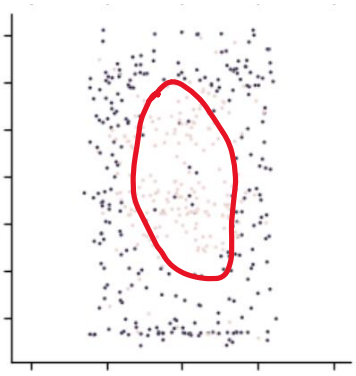


Q16

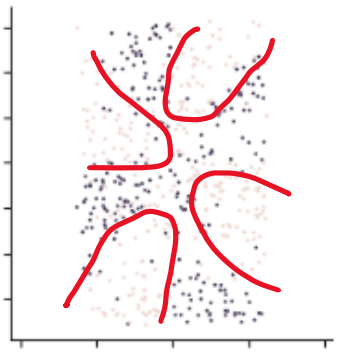
We found that the PCR_02 and PCR_06 features are helpful for predicting the risk target variable, since we can see that when the data is split according to the SpecialProperty and we use the mentioned features as our datapoints we get that the risk is separable, as can be seen below.

The graphs below show PCR_02 in the x axis and PCR_06 in the y axis, and we can see from the added lines that the data is pretty separable when we split the data by the SpecialProperty

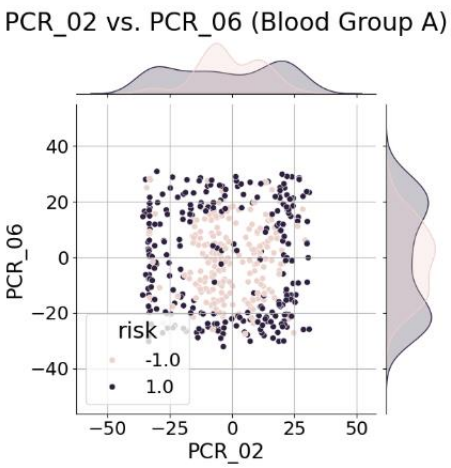
O+,B+

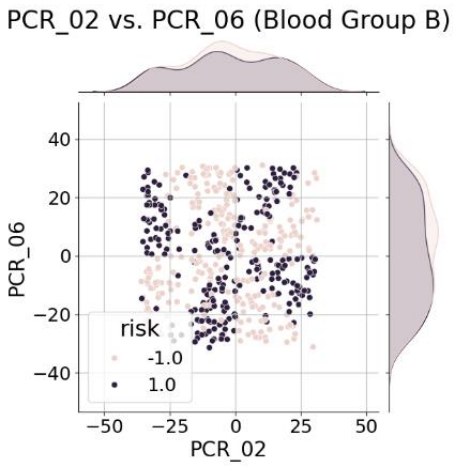


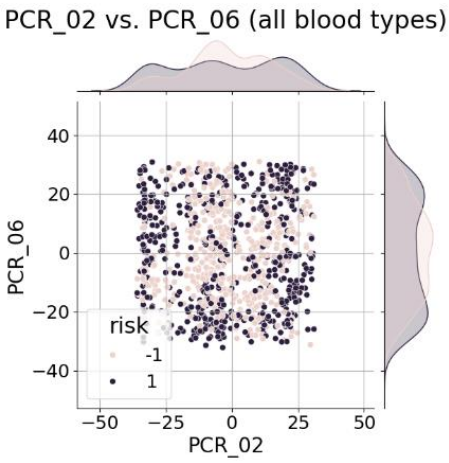
The rest



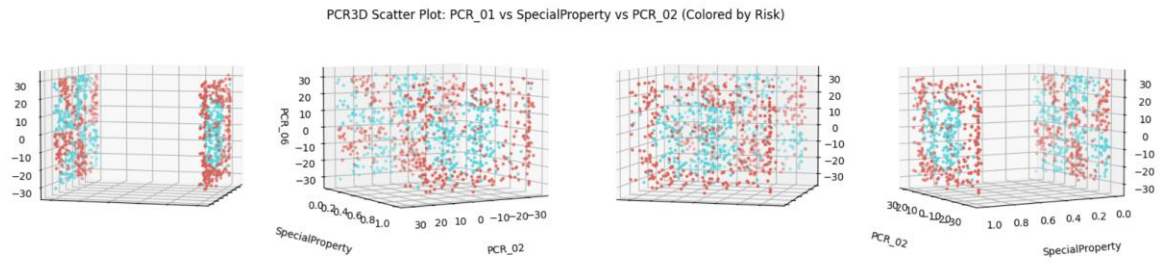
Q17







Q18



Q19

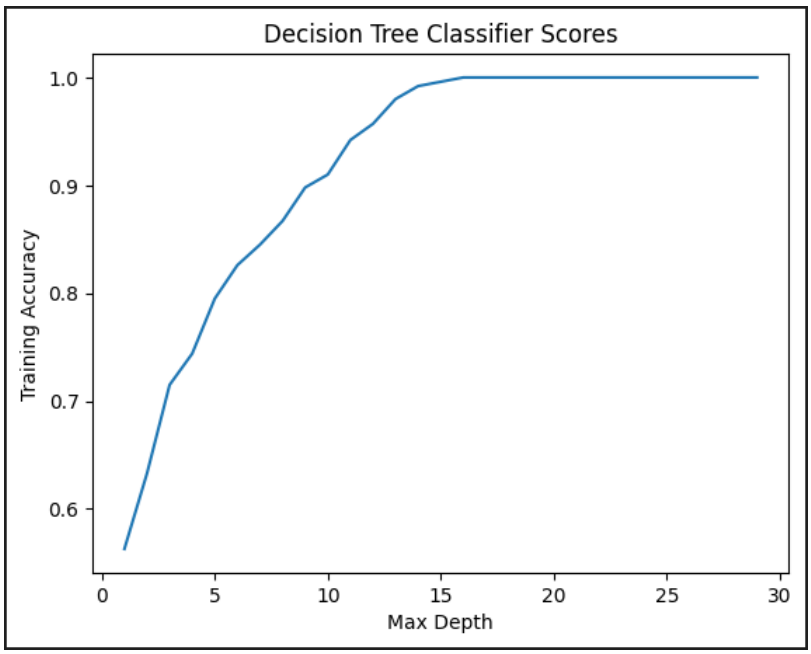
A decision tree of max-depth=3 won't fit the data well, as it won't be able to fully separate the datapoint into target risk = {+1,-1} because it can make at most three splits which will give us underfitting, as out datapoints are 3-dimensional and the pattern we have isn't simple and it requires more splits to be separated.

As we can see in the graph below a max depth of 3 will give us a training accuracy of around 0.7 which isn't good.

Q20

As opposed to the previous question a decision tree with max-depth=30 would get us great result in terms of training accuracy, as we'd have a very complex tree that could fit each pattern we found in the datapoints which will give us perfect training accuracy, and would lead to overfitting as now we have enough decision room to isolate even the noisy datapoints, but this isn't bad in terms of fitting the training data and that's what were asked about here.

To support our claims we've plotted the following graph that shows us the training accuracy as a function of the max-depth parameter we pass to the decision tree model.



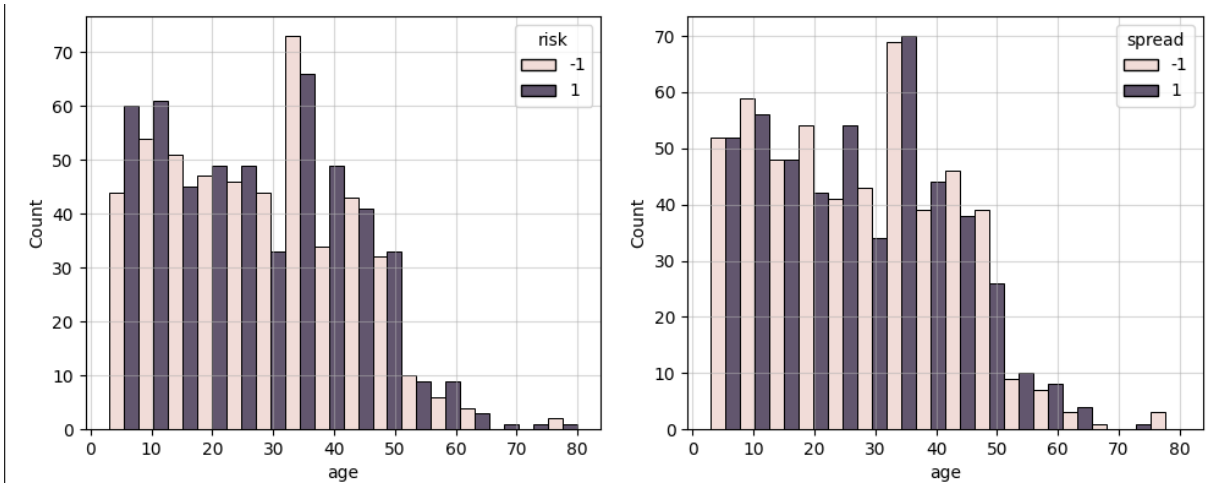
Q21

Looking at the data we can see that using 1-NN won't fit the our data points that well, as we have many Imposters that will be mostly misclassified by our model, secondly we can see that there isn't really a margin between "regions" of different classifications which is why points that are on the edge of a certain classification region with 1-NN would be misclassified although intuitively we can see a definitive border between the regions.

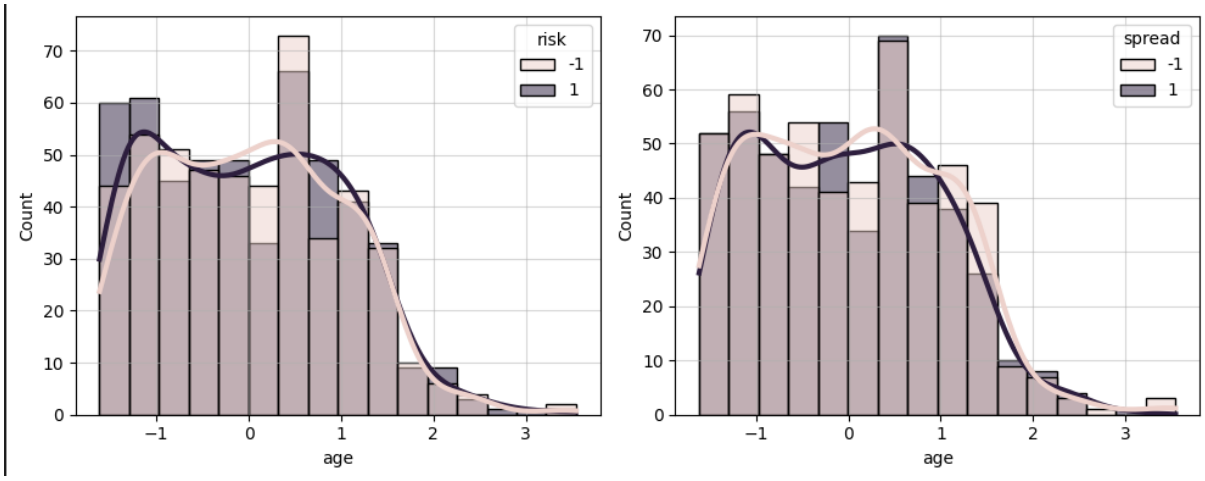
Finally and most importantly the reason why K-NN model won't fit our data well is that our SpecialProperty feature is on a different scale than PCR_02 and PCR_06 and in a distance based model that could largely affect our accuracy in a way that a point that's in blood group A might find it's closest neighbor in blood group B because of the difference in scale between our axis, which could cause the model to define wrong "decision regions".

Q22

Before:



After:



Q23

Normalizing or scaling our data will not impact the results of questions Q19 and Q20, as decision trees inherently exhibit insensitivity to the scale of features. The splitting criteria at each node are based on relative feature values rather than their absolute magnitudes. Consequently, the model's structure and predictions remain robust and unaffected by changes in the scale of input features.

On the other hand, for distance-based models, normalization has a substantial impact on the model's accuracy and decision-making. In the case of Q21 and our 1-NN model, normalizing the data introduces a notable difference in distances between points in blood group A and blood group B. This results in more meaningful and well-defined classification regions, enhancing the overall performance and accuracy of the model.

Q24

(i) Rule of thumb for choosing standard normalization (we decide by looking at univariate analysis):

1. Follows a normal distribution or is approximately normally distributed.
2. Contains outliers that you want to handle robustly.

(ii) For any other features that have values outside of the [0,1] range we used minmax normalization.

Feature Name	**Keep**	**New**	**Normalization method**	**Explanation**
Patient_id	X	X		Removed because it isn't relevant for our classification
Age	V	X	standard	By (i)
Sex	X	X		Dropped as we converted this feature to an equivalent numeric
isMale	V	V		The numeric version of the sex feature.
Weight	V	X	standard	By (i)
Blood_type	X	X		Dropped as were now using the SpecialProperty numeric feature instead
SpecialProperty	V	V		Added following the HW instructions
Current_location	X	X		Was replaced by Current_location_x

				and Current_location_y
Current_location_x	V	V	standard	By (i) + Added as a numeric version of current_location
Current_location_y	V	V	standard	By (i) + Added as a numeric version of current_location
Num_of_siblings	V	X	Standard	By (i)
Happiness_score	V	X	Standard	By (i)
Household_income	V	X	Standard	By (i)
Conversation_per_day	V	X	Standard	By (i)
Sugar_levels	V	X	Standard	By (i)
Sport_activity	V	X	Standard	By (i)
Symptoms	X	X		Was replaced by a feature for each symptom
Cough	V	V		Added to have a numeric representation to having this symptom
Fever	V	V		Same as above
Shortness_of_breath	V	V		Same as above
Smell_loss	V	V		Same as above
Sore_throat	V	V		Same as above
Pcr_date	X	X		Dropped because we think its not relevant.
PCR_01	V	X	Min_max	By (ii)
PCR_02	V	X	Min_max	By (ii)
PCR_03	V	X	Min_max	By (ii)
PCR_04	V	X	Standard	By (i)
PCR_05	V	X	Min_max	By (ii)
PCR_06	V	X	Min_max	By (ii)
PCR_07	V	X	Standard	By (i)
PCR_08	V	X	Standard	By (i)
PCR_09	V	X	Standard	By (i)
PCR_10	V	X	Standard	By (i)